MTH 202 Chapter 5: Topics In Vector Calculus

5.2.1 Line Integrals Worksheet

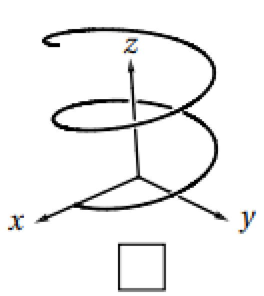
1. Let C be the curve in $\mathbb{R}^3$ parameterized by $\mathbf{r}(t) = \langle \sin(t), 2t, \cos(t) \rangle$ for $0 \le t \le \frac{\pi}{2}$.

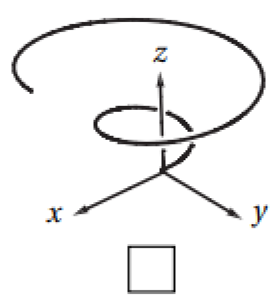
a) Evaluate the integral $\int_C x^2 z\, ds$.

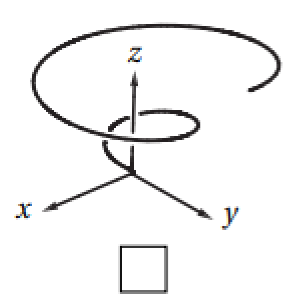
b) Evaluate the integral $\int_C x^2 z\, dx$.

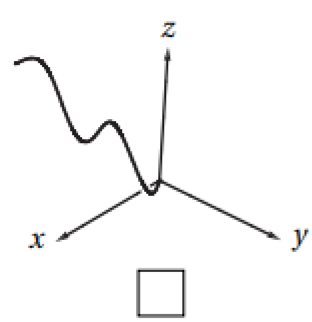
c) Evaluate the integral $\int_C x^2 z\, dy$.

2. Mark the picture of the curve in $\mathbb{R}^3$ parameterized by $\mathbf{r}(t) = \langle t\cos(t), t\sin(t), t \rangle$ for $0 \le t \le 4\pi$.









3. Let C be the curve parameterized by $\mathbf{r}(t) = \langle \cos(t), \sin(t), 2t \rangle$ for $0 \le t \le 2\pi$.

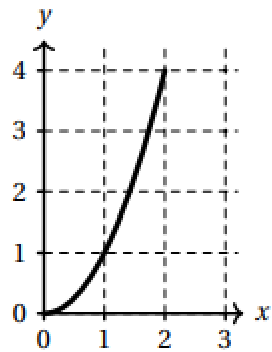
a) Compute $\int_C z\, ds$.

b) Compute $\int_{-C} z\, ds$.

4. Find the mass of a thin wire in the shape of the curve $\mathbf{r}(t) = \left\langle \sin(t), 2\sin(t), \sqrt{5}\cos(t) \right\rangle$ where $0 \le t \le \pi$, if the wire has density function $\rho(x, y, z) = x$.

5. Suppose that $\mathbf{r}:[0,2]\to\mathbb{R}^2$ is a parametric curve in the plane and that $\mathbf{r}(t)$ and $\mathbf{r}'(t)$ have the values given in the table at left. Check the box below the picture that could be a plot of $\mathbf{r}(t)$.

t	$\mathbf{r}(t)$	$\mathbf{r}'(t)$
0	$(0,0)$	$\mathbf{i}$
1	$(1,1)$	$-\mathbf{i}$
2	$(2,4)$	$\mathbf{i}+4\mathbf{j}$



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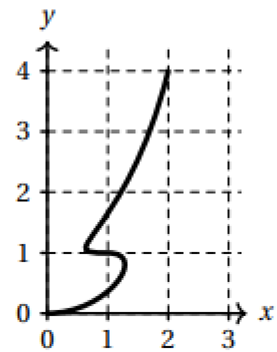

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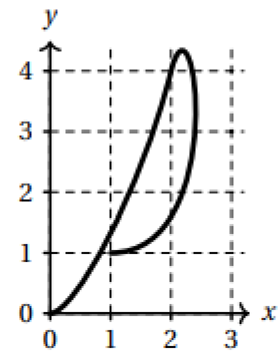

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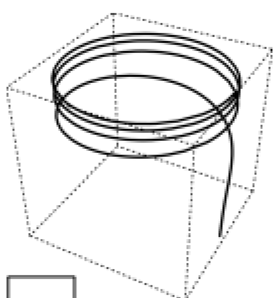
6. Let C be the oriented curve parameterized by $\mathbf{r}(t)=\langle\cos(t),\sin(t),e^t\rangle$ where $0\le t\le 8\pi$.

a) Check the box next to the picture which best matches C.



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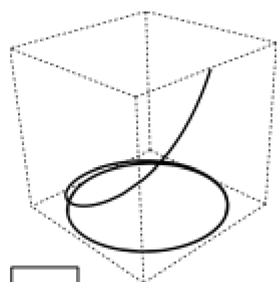

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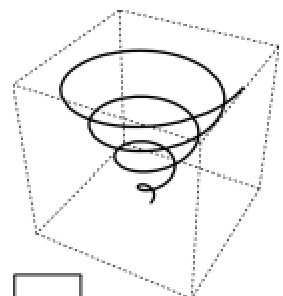

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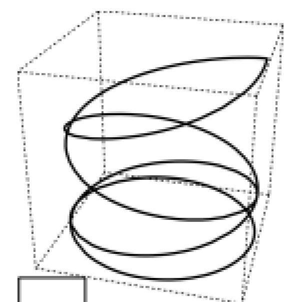

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b) Calculate the line integral $\int_C z\,dz$

c) Calculate the line integral $\int_{-C} z\,dz$.

7. Suppose a wire can be modeled by the segment of the circle $x^2+y^2=4$ from $(2,0)$ to $(0,2)$, when moving counterclockwise. If the linear density of this wire is measured by $f(x,y)=xy$ for any point on the wire, then set up an integral which represents the mass of this wire.

8. Let $f(x,y,z)=xy-xz$, and C be a curve which is the intersection of the surfaces $y=x^2+1$ and $z=x^2-1$ from $(0,1,-1)$ to $(1,2,0)$. Compute the line integral $\int_C f(x,y,z)\,ds$.

9. Evaluate the line integral $\int_C e^x\,dx$, where C is the arc of the curve $x=y^3$ from $(-1,-1)$ to $(1,1)$.

10. Consider the line integral $\int_C xye^{yz}\,dy$, where C is the arc of the curve parameterized by $\mathbf{r}(t)=\langle t,t^2,t^3\rangle$ from $0\le t\le 1$.

a) Evaluate $\int_C xye^{yz}\,dy$.

b) Evaluate $\int_{-C} xye^{yz}\,dy$.

11. Evaluate the line integral $\int_C x^2dx+y^2dy$, where C consists of the arc of the circle $x^2+y^2=4$ from $(2,0)$ to $(0,2)$ followed by the line segment from $(0,2)$ to $(4,3)$.

12. Evaluate the line integral $\int_C (y+z)\,dx+(x+z)\,dy+(x+y)\,dz$, where C consists of the line segments from $(0,0,0)$ to $(1,0,1)$ and from $(1,0,1)$ to $(0,1,2)$.